How Task-Assignment Notifications Work

TeamTask · brief notes

In short

When someone assigns a task to a user, the backend saves an in-app notification for that user in MongoDB. The frontend checks for unread notifications every 30 seconds and shows the count on the bell icon. The user opens the Notifications page to read them and mark them as read.

Step by step

1. **A task gets an assignee.** This happens when a task is created with an assignee (`POST /api/tasks`) or when the assignee of an existing task is changed (`PATCH /api/tasks/:id`).
2. **The backend saves the task** and writes an *assigned* entry to the audit log.
3. **It calls** `notify()` (backend/src/service/notificationService.js) with the new assignee, the person who assigned it, the type *assigned*, and a message such as *“Priya assigned you ‘Fix login bug’”*.
4. `notify()` **runs two checks** and skips the notification if either one is true:
   • the assignee is the same person who assigned it (you don't get notified about your own action);
   • the assignee turned off *“a task is assigned to me”* in Settings.
5. **Otherwise it saves a Notification** in MongoDB with the recipient, type, message, a link to the task, and *read = false*.
6. **The frontend picks it up.** Every 30 seconds, and on every page change, it calls `GET /api/notifications?unread=true`. The unread count appears as a red badge on the bell and next to Notifications in the sidebar.
7. **The user reads it.** The Notifications page lists the latest 50, newest first. Each one links to its task. The user can mark one as read or mark them all as read.

API endpoints

Method	Endpoint	What it does
`GET`	`/api/notifications`	Latest 50 notifications and the unread count (?unread=true for unread only)
`PATCH`	`/api/notifications/:id/read`	Marks one notification as read
`PATCH`	`/api/notifications/read-all`	Marks all of the user's notifications as read
`PATCH`	`/api/auth/preferences`	Turns the assigned / updated notifications on or off

Every endpoint needs the user to be logged in. Users can only see and change their own notifications.

Good to know

- **Not instant:** a new notification can take up to 30 seconds to show on the bell, or shows as soon as the user moves to another page.
- **In-app only:** no email or push notifications are sent.
- **Reassigning:** only the new assignee is notified. The previous assignee is not told the task was taken away.
- **Unassigning** a task (setting it to Unassigned) sends no notification.
- **Task updates** are a separate type, *updated*. When a task's other fields change, its creator (and its assignee, if the assignee didn't change) get an “updated” notification, with the same two checks.